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CREATIVITY IN DESIGN: EXPERIMENTING AND INNOVATING AT TEAMLAB JAPAN

The office building, spanning over four floors, in a quiet street of bustling Tokyo was like few others. Entering the first floor on a cool spring day in May 2015, Toshiyuki Inoko, the founder and chief executive officer of the Japanese digital-artist collaborative teamLab, looked around the room with satisfaction. In a large open space with no room dividers, he could see almost a hundred artists, architects, mathematicians, web and print graphic designers, programmers and engineers busy at work. Large tables, also doubling as whiteboards, were being used to create new designs. Groups were huddled around another set of tables to collaborate and brainstorm ideas. On the walls, there were animated holograms showcasing several of the innovative digital art pieces that the teamLab team had developed. Inoko thought to himself that it had been exactly 14 years since he had founded teamLab with the idea of creating art using digital technology, and the model finally seemed to be coming together. From a small group of five, struggling to obtain a single project, the company had grown to over 400 employees working on more than one hundred projects at any point in time.

In 2001, while still at university, Inoko set up teamLab, Inc. with four of his friends. At that time, the internet was becoming increasingly popular, and, recognizing its potential to change the world, he decided to use that technology to create art based on digital ideas. As he elaborated, “I thought it was time to use the Internet and develop some creative digital art for this new technology, which could change the values and thinking of the new generation.”

However, as art in digital form was a concept not easily understood or accepted by conventional society, Inoko and his team struggled for almost ten years to build a strong customer base with a robust project pipeline. In the past four years though, projects had started streaming in, and teamLab had since displayed a range of products combining aesthetically advanced art and cutting-edge technology, which were exhibited across the globe.

Despite such success, Inoko realised that a different set of challenges had arisen. teamLab had grown dramatically, and it was critical to ensure that the passion and creative edge of its founders, along with a spirit of collaboration and co-creation among team members, were maintained. Moreover, given the increasing scale of business, it was becoming necessary to bring in some structure and a system to orchestrate what was, fundamentally, a chaotic creative process. The worry was that such a structure could also potentially dim the creative spark. How could Inoko successfully manage this apparent paradox of creativity and business discipline?

Havovi Joshi prepared this case under the supervision of Professor Ali Farhoomand for class discussion. This case is not intended to show effective or ineffective handling of decision or business processes.

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The Idea that Ignited teamLab

In the mid-1990s, Inoko was studying engineering at the University of Tokyo. At that time, the internet was taking off, and Inoko quickly recognized its potential power. He began to think about moving beyond developing the technology itself to understanding how digital art and technology could be integrated. Over the next few years, he exchanged ideas with his friends, and, in March 2001, five of them got together to establish teamLab. Commenting on the thinking that went behind establishing teamLab, Inoko explained:

The main purpose of launching teamLab is to fulfil my personal interest. Not by myself, but with people with various specialties enabling us to experiment with a lot of new things. I wanted to offer a place where we can create and learn something new through experimentation. That's why my company's name includes the word "Lab." Right before we started college, the Internet came out and I recognised the coming of a new digital society, which led me to have an interest in creating something new in that new society. I was especially interested in changing people's sense of value by innovation through technology and the arts.

- Toshiyuki Inoko, founder and chief executive officer, teamLab¹

Recalling those early days, Daisuke Sakai, one of the co-founders of teamLab, said,

When we started 14 years ago, we were just five friends, all engineers – three from the University of Tokyo and two from the Mechanical University of Tokyo. Inoko always wanted to start a company, and although we didn't know how to go about it, we just went ahead, rented some rooms and started off. But we realized that we could not compete with all the great technological innovation coming out of Silicon Valley. So we decided to focus on expanding the meaning of art using digital technology.

- Daisuke Sakai, co-founder of teamLab²

The first project that the newly founded company worked on was installing some digital art at a shop of the cosmetics company Shiseido in Ginza, Tokyo. While this did not make much money, it did help spread the word. However, new projects started trickling in very slowly. As Sakai explained, “Since we were all just engineers, we knew how to make creative and innovative things – but we didn't know how to sell and we didn't know how to make a company.”³

After struggling for almost a decade, teamLab had a breakthrough in 2011, when they were provided an opportunity to display at an exhibition in a Taiwanese gallery called Kaikai Kiki Taipei. This heralded the company's formal entry into the international art market, which immediately provided fresh impetus to the lagging business. In 2011, the group also participated in the 14th Annual Japan Media Arts Festival, and thereafter in the Kagoshima Art Festa in 2013.⁴ Some of the large exhibitions its products were displayed in included “We Are the Future” at the National Museum of Fine Arts of Taiwan in 2012, and “SUPER RELAX” at Japan's Saga

¹ Suno, R. and Hamaker, S. (24 October 2011) “An interview with Toshiyuki Inoko”, J-COLLABO.org, <http://j-collabo.org/interviews/toshiyuki-inoko/> (accessed 3 August 2015).

² In conversation with the case authors on 9 June 2015.

³ Ibid.

⁴ Singapore Biennale (2013) “If the world changed”, http://www.singaporebiennale.org/downloads/folios/teamLab_Final.pdf, (accessed 12 August 2015).

Castle History Museum the same year. By mid-2015, about half of teamLab's art and installation business came from overseas.

Designing Digital Creations

Referring to themselves as “ultra-technologists,” teamLab's aim was to achieve a balance between art, science, technology and creativity. The company produced a range of digital technology products. Inoko elucidated:

I loved the idea of technology changing the world and art changing people's minds. So I committed myself to digital technology, which was a new field and decided to create art based on digital ideas.... We wanted to create digital artworks that were innovative, and changed people's minds and values. I believe that, because it didn't conform to society, there are intellectual or cultural ideas and knowledge that have been abandoned over time. Using them through our work, we want to provide an insight into these ideas and suggest an alternative way to live in contemporary society.

- Toshiyuki Inoko, founder and chief executive officer, teamLab⁵

In the early 2000s, digital creations were a relatively unknown concept, and teamLab was one of the pioneers in this space. Its work involved the use of extremely sophisticated digital media that could be presented via computers, high definition monitors, and projections based on the space available for installation. The use of technology and interactive display was, in fact, a key factor driving teamLab's design philosophy.

In essence, teamLab believed that producing digital works could extend the beauty of an artwork itself, as it enabled a completely different experience, allowing the spectator to become a part of the artwork. The artworks produced by teamLab were thus typically interactive. Providing an example to elaborate on the concept, Sakai explained:

We exhibited two interactive, digital art pieces in the Japan pavilion at Expo Milan 2015. The theme of the Expo was Japanese food and culture, and we wanted to make the audience feel like they were in an actual rice field. So in one room, there was a projection space where visitors walked through a space that has screens of various heights displaying rice fields. It actually responded to their presence, and ripples. It also reacted to their behavior, so if you tried to touch a fish, it would move away. In the other room, information on Japanese food and culture was provided through a digital waterfall. And the visitors could transfer the images that flowed down the waterfall, along with some information on it, onto their smartphones by simply touching the image. So it was something more tactile – it was experiential and immersive.

- Daisuke Sakai, co-founder of teamLab⁶

Inoko explained a key difference between physical and digital art, which underpinned the immense potential to be explored using digital art:

⁵ Dickie, A. (13 January 2014) “A conversation with Toshiyuki Inoko”, OCULA Conversation, <http://ocula.com/magazine/conversations/toshiyuki-inoko/> (accessed 3 August 2015).

⁶ In conversation with the case authors on 9 June 2015.

In the past, information came with some kind of physical medium as a set. For example, information such as music came in the form of a vinyl record. Today, thanks to digital technology, music exists independent of the vinyl record. In short, information has been freed from the limits of material media. This is the same in artistic expression. Art is being freed from the limits of its conventional media such as a canvas. This means that digital art has no physical boundary and can contain so much information... This change is happening very quickly and artworks are ever more flexible and able to change via the viewers' participation. In a way, this has become the norm.

- Toshiyuki Inoko, founder and chief executive officer, teamLab⁷

Digital art enabled artists to have far more autonomy and freedom with regard to the space where the artwork is to be installed. Artists could manipulate and use much larger amounts of space, allowing viewers to experience artworks more directly.⁸ For instance, in April 2015, at a large display in Tokyo's National Museum of Emerging Science and Innovation entitled "Floating Flower Garden," 2,300 living flowers were suspended from the ceiling. These flowers would not only rise and fall around visitors as they moved through the show—sensors would detect an approaching viewer and the flowers in that viewer's vicinity would rise—but their scents, too, would change throughout the day. It was believed that this immersive, multi-sensory interaction would provide visitors an opportunity to see nature in a more personal way.

A typical project, such as the one described above, would usually take three to six months to complete. Some of the larger projects could take up to a year. For instance, a project at Skytree, which was the highest tower in Tokyo, was about 40 metres high and, while it was all-digital, it was composed of handwritten elements and had about 35,000 objects in the painting which moved.

teamLab's clients were typically corporate customers. The company's works were also frequently displayed in exhibitions and museums. In addition to digital artwork, teamLab also developed new innovations in technology by producing products, phone applications and digital programs for commercial use.⁹

Technology Serves Art: Creating a Successful Business Model

In Inoko's opinion, for the business to succeed, it must develop products that "look like art." He believed that customers were becoming increasingly sophisticated, and would expect and demand an "elegant, artistic" experience when they purchased the company's product or service.¹⁰ For this to happen, technology would have to serve art, and not vice-versa. This thinking served teamLab well, as the technology being used was not particularly hi-tech or unique. What was highly innovative though, and also teamLab's key strength, was the use of technology to create an altogether new experience.

Sakai elucidated the creative process further:

⁷ Japan Society (12 June 2014) "Garden of Unearthly Delights", <http://www.gardenofunearthlydelightsexhibition.com/home/teamLab/>, (accessed 18 July 2015).

⁸ Digital meets Art (6 August 2015) "About teamLab, Tokyo-based digital artist collaborative", <http://www.digitalmeetsculture.net/article/teamLab-tokyo-based-digital-artist-collaborative/> (accessed 6 August 2015).

⁹ Singapore Biennale 2013, "If the world changed", http://www.singaporebiennale.org/downloads/folios/teamLab_Final.pdf, (accessed 12 August 2015).

¹⁰ Business Times (7 September 2012) <http://www.team-lab.net/en/article/btimes0907.html> (accessed 18 July 2015).

When we start generating ideas about a project, it is in discussion with all our people – engineers, designers and the project managers. We don't start our thinking from a technology perspective – instead we first think about what feels good, and the experience of the target audience, that is the person who will experience that. So we use a lot of senses. And we do use new technology to make it happen - but that is not the purpose. We are very goal-driven to make the people feel the way we want to make them feel. So we are not like companies who say, "I have this technology, how should I use this?" and instead, we say, "How can I plan this project to make this experience happen?" That's the way we create things by evolving and adapting our past ideas to make it better.

- Daisuke Sakai, co-founder of teamLab¹¹

This approach became teamLab's competitive advantage, as it was much harder for a competitor to copy or imitate an "experience" than a new technology. In Sakai's view, there was no other company that had teamLab's scale in the business of art technology. He also added, "We don't have a name for what we are producing, but when I want to sell you our product, I will get you there and make you experience it – as it's the experience that counts, and is very different seen on video."¹²

It was precisely that difficulty in pigeonholing teamLab's value proposition – which came from being an undefined mix of exhibitor, designer, engineer and artist - that encouraged and enabled its people to be creative and innovative.

A Flat Organizational Structure

teamLab's employee base was a young team, with an average age of around 30. Recognizing the importance of a diverse mix in ensuring creativity, the team was made up of several nationalities, including members from China, Korea, France and Brazil. The company prided itself on keeping its organizational structure flat, with minimal hierarchy. As Sakai explained:

We don't have any levels, and we think that makes it easier for people to collaborate and interact with each other; and understand how they think, what to make, what is good, and what is creative. In this office, we don't have any partitions, and we do that on purpose because if there's a door, you have to knock and may hesitate to go in. But we want to make every single communication easy, so that people want to talk to each other.

- Daisuke Sakai, co-founder of teamLab¹³

The compensation package reflected this philosophy, and a critical component of evaluating a team member's performance on a project was their ability to work in a team, motivate peers and have a collaborative attitude in the workplace.

Not an Engineer or a Designer – but a "Catalyst"

Sixty percent of teamLab's employees were engineers, such as smartphone engineers, web engineers and database engineers, and another 20 percent were creative designers, including graphic designers, web designers, and art technicians. Only three percent of the team were employed in back-office operations. The remaining employees were designated as "Catalysts." These people were responsible for all activities other than designing or programming. They

¹¹ In conversation with the case authors on 9 June 2015.

¹² Ibid.

¹³ Ibid.

would act as project managers, in charge of gathering all the information for the project, disseminating it to the team members, managing the costing and pricing, etc.

Elaborating on the origin and importance of a Catalyst's role, Sakai said:

We thought we could make, design and do anything and everything because we were engineers and had designers. But then a lot of projects went wrong and failed, because we couldn't manage them, and we didn't know how to connect with designers. You see, we had never worked anywhere else earlier, and so we didn't even know what directors or project managers do. But we have now learnt that there must be such a role – not a level - but a role to connect all the aspects of the project to make it happen.

- Daisuke Sakai, co-founder of teamLab¹⁴

The Catalysts were thus key players contributing to teamLab's success at delivering projects. While some were technology-savvy, there were yet others who were not, but knew how to work well with the customers, designers and programmers.

Co-creation and Incremental Innovation Arising from Collaboration

The creative process at teamLab was highly collaborative. This was the key reason that, despite having small offices in Shanghai and Taipei, all the creative work was done in the Tokyo office.

Elaborating on the importance of such collaboration, Takashi Kudo, Chief Brand Officer, teamLab, concluded:

We craft things as a team. New ideas tend to arise from the cross-pollination of technical insights from all kinds of different areas. One person can't possibly do all the thinking. You also can't have non-technical people coming up with the ideas, because without an understanding of existing technology, you won't know what's currently possible with it. That's why, at teamLab, we encourage constant communication between members, and it's through our process of contributing ideas, taking action, and making prototypes together, as well as brainstorming together, that we provide our products and services.

- Takashi Kudo, chief brand officer and editor, teamLab¹⁵

Sakai added, "We are working on a number of projects, but the people will move from project to project, so it is like copying the DNA to each other.... The only type of people who should be here are those who believe that they can do more in a team – because if they want to be an artist working alone, they shouldn't be here."

To further enhance the sharing of knowledge that would drive the collaborative effort of co-creation, teamLab made all projects available to every employee. It had also implemented a chat system technology, which could be viewed by project or person. Sakai explained:

¹⁴ Ibid.

¹⁵ Martin, R. (28 November 2013) "TeamLab's vision of Christmas future: A 2-ton, 4D animated tree in Japan", *The Bridge*, <http://thebridge.jp/en/2013/11/team-lab-christmass-tree> (accessed 15 July 2015).

For any one project, everyone involved has a different background, and has been assigned to different projects. But there is a project database and manager for each and every thing - and they can go and find an earlier project that relates to this new project – that way they have some idea that could be referenced from before. Then they add to it to create a new idea, and multiply it to make it a creative new idea. So the one thing that is very important is to stay in one place to have access to these ideas. It also helps the people to share new ideas with each other, or just consult.

- Daisuke Sakai, co-founder of teamLab¹⁶

This system had an added advantage in that it enabled the training of new recruits, who were assigned a mentor to guide them through the database and the ideas stored within.

Creating and Marketing a Strong Brand

For almost a decade after being founded, teamLab's artistic works were snubbed in Japan. This could also be attributed to the fact that art going digital had been a slow-moving process. One reason that digital art had not done well was because of challenges faced in terms of poor resolution and clunky interface. This issue had been largely mitigated, and the industry had been beginning to take off, as reported by an advocate of the digital arts community:

A raft of digital art start-ups like ArtLion, ArtKick and Artcast had built libraries of high-resolution fine art and photography, and the ease of changing art on them means you can have a different piece for any occasion; Picasso at dinnertime or Mondrian in the morning.... Communities of digital artists like DeviantArt and NewHive have exploded, providing increasing amounts of high-quality digital art created to be viewed on a screen.... It's taken a long time for the art world to go digital, but new encryption technologies like Monegraph are offering ways for collectors to mitigate piracy, and authenticate the provenance of a work. Digital artists like Rafaël Rozendaal are experimenting with new models for selling public work privately, and as artists become more comfortable with the medium (and start actually making money from it), the industry is likely to take notice.

- Kim Gordon, co-founder and chief executive officer at Depict.com¹⁷

Kudo had joined teamLab in 2011, and began the process of creating the company's brand in earnest. Commenting about the branding process, he said:

¹⁶ Ibid.

¹⁷ Gordon, K. (8 February 2015) "The Year That Tech Needed the Art Industry", *Crunch Network*, <http://techcrunch.com/2015/02/08/the-year-that-tech-needed-the-art-industry/> (accessed 20 August 2015).

I have to put the word out about our brand, and with the Internet, we could reach out to everybody - so it's the first priority for us. But then we thought about what information we should put on the Internet. And I believe that people prefer to have the information explained to them, rather than just getting details listed on a website. I also think that teamLab's fundamental business is that we are creative – and as it is fun stuff, people are going to talk about it and share it in any case - so we should focus on what we do. Hence, if we make some new product or deliver a new service, we should explain everything about it in 130 words. Then people can talk and tweet about it. So I try to explain our work in a very pure elemental way, and of course it helps that it is so visual... For five years now, we have been making our website, and we archive all that we do on it. That helps our branding, as I just give our link, and everyone can understand what we have to talk about.

- Takashi Kudo, Chief Brand Officer and Editor, teamLab¹⁸

Given that teamLab did not believe in having a marketing team, the strong presence that it had on the internet was particularly advantageous. Moreover, as almost all the projects that the company had completed had been featured on websites and on YouTube, which people shared via social media, its strategy for branding was further reinforced, and teamLab had been quite successful in getting more project requests through the internet. In the past five years or so, teamLab had also received considerable media attention.

The Road Ahead

We have grown from five to 400 people. The more we have grown, the more things we have developed and the more things that we can do. So the depths of our creativity have expanded... Our dream is to make our brand larger, to make ourselves bigger, and to produce even more impressive and surprising artwork to show to the world.

- Daisuke Sakai, co-founder of teamLab¹⁹

teamLab's business model appeared to have successfully harnessed the creativity and expertise of its people to deliver innovative and imaginative products that were becoming increasingly popular in the market. While this space was still very new and had no other established players, teamLab would have to ensure that they did not let the excitement and novelty of their products wear off, as it was this that provided them with a strong competitive edge.

Moreover, with growth and success came a different set of challenges. The team had grown, and while that brought new experiences and a heady mix of diverse personalities to the table, it also meant that collaboration among the team members would have to be managed more systematically. Systems and processes would need to be brought in to manage the growing business effectively, but this could also potentially impede an intrinsically disorderly creative process.

Looking ahead, how could Inoko successfully manage the spectacular growth of his company and the challenges that it brought with it, while ensuring that the creative spark continued to burn bright?

¹⁸ Martin, R. (28 November 2013) "TeamLab's vision of Christmas future: A 2-ton, 4D animated tree in Japan", *The Bridge*, <http://thebridge.jp/en/2013/11/team-lab-christmass-tree> (accessed 15 July 2015).

¹⁹ In conversation with the case authors on 9 June 2015.